

Computable and uncomputable functions

Exercise sheet 1

Exercise 1.1. A total function $f: \mathbb{N} \rightarrow \mathbb{N}$ is called *non-increasing* if $f(n+1) \leq f(n)$ for all $n \in \mathbb{N}$. Prove that every non-increasing function is computable.

Exercise 1.2. Let $f: \mathbb{N} \rightarrow \{0, 1\}$ be such that $f(n) = 1$ if the decimal expansion of π contains n consecutive 9s, and $f(n) = 0$ otherwise. Is f computable?